

1 PageRank

Consider the provided raw adjacency matrix of a graph:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & 1 \\ 0 & 0 & \frac{1}{2} & 0 \end{bmatrix}$$

1. Convert A into a column-stochastic matrix M (W1.2 - Slide 18).

2. As discussed in the lecture, PageRank computes the page rank vector r as

$$r = M \cdot r$$

(W1.2 - Slide 20) where r is the eigenvector of the column-stochastic M corresponding to eigenvalue $\lambda = 1$. Compute the PageRank of each node in the above graph.

3. What is link spam? Explain how link spam increases the PageRank of a target page (W1.3 - Slide 18 to 21).

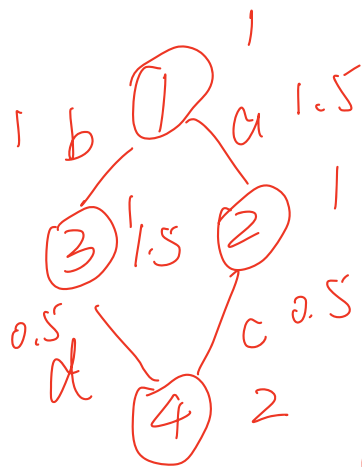
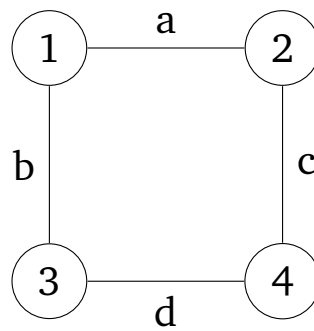
$$\begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & 1 \\ 0 & 0 & \frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{bmatrix}$$

$$\begin{cases} \frac{1}{2}r_2 = r_1 \\ \frac{1}{2}r_1 + \frac{1}{2}r_3 = r_2 \\ \frac{1}{2}r_1 + \frac{1}{2}r_2 + r_4 = r_3 \\ \frac{1}{2}r_3 = r_4 \end{cases} \Rightarrow$$

$$r_1 + r_2 + r_3 + r_4 = 1$$

2 Edge Betweenness

Consider the provided graph, and compute the edge betweenness of the edges using the Brandes' algorithm (W2.1 – Slide 12 to 16).



Handwritten calculations for edge betweenness:

$$EB(a) = 1.5$$

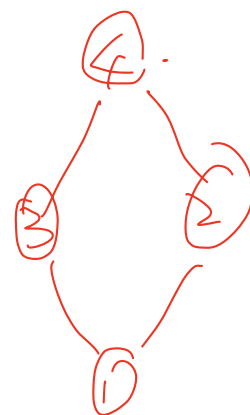
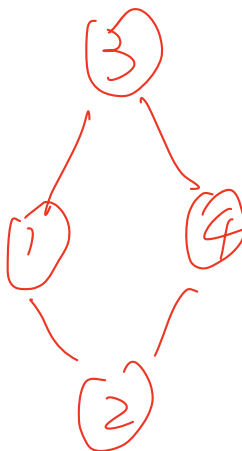
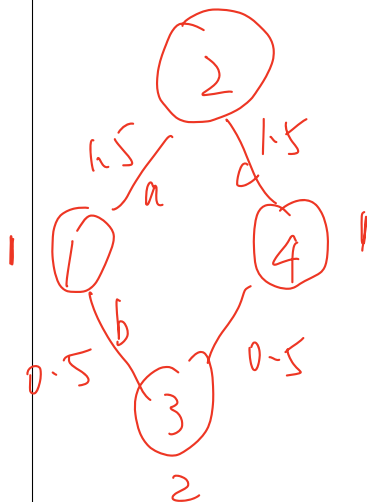
$$EB(b) = 1.5$$

$$EB(d) = 0.5$$

$$EB(c) = 0.5$$

Handwritten calculation:

$$1 \times \frac{1}{2} = \frac{1}{2}$$



Handwritten formula for edge betweenness:

$$EB(a) = \frac{\text{回该图的 } EB(a) \text{ 的合}}{2}$$

Handwritten note: 算了两遍

3 Spectral Clustering

Consider the provided raw adjacency matrix of a graph:

$$W = \begin{bmatrix} 1 & 0.2 & 0 & 0 \\ 0.2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0.5 \\ 0 & 0 & 0.5 & 1 \end{bmatrix}.$$

1. Compute the Laplacian matrix $L = D - W$ (W2.2 - Slide 14).
2. Apply the eigen-equation $|\lambda I - L| = 0$ to obtain the eigenvalues of the Laplacian matrix (you may need to compute the determinant¹).
3. Use the second eigenvalue λ_2 to find the best partition of the graph into two communities (W2.2 Slide 17).

(1)

$$D: \begin{bmatrix} 1.2 & 0 & 0 & 0 \\ 0 & 1.2 & 0 & 0 \\ 0 & 0 & 1.5 & 0 \\ 0 & 0 & 0 & 1.5 \end{bmatrix}$$

$$L: \begin{bmatrix} 0.2 & -0.2 & 0 & 0 \\ -0.2 & 0.2 & 0 & 0 \\ 0 & 0 & 0.5 & -0.5 \\ 0 & 0 & -0.5 & 0.5 \end{bmatrix}$$

(2)

$$I = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

$$\lambda I = \begin{bmatrix} \lambda & & & \\ & \lambda & & \\ & & \lambda & \\ & & & \lambda \end{bmatrix}$$

$$|\lambda I - L| = 0$$

$$= \begin{vmatrix} \lambda - 0.2 & 0.2 & 0 & 0 \\ 0.2 & \lambda - 0.2 & 0 & 0 \\ 0 & 0 & \lambda - 0.5 & 0.5 \\ 0 & 0 & 0.5 & \lambda - 0.5 \end{vmatrix} = 0$$

for λ

$$= \begin{vmatrix} \lambda - 0.2 & 0.2 \\ 0.2 & \lambda - 0.2 \end{vmatrix} \cdot \begin{vmatrix} \lambda - 0.5 & 0.5 \\ 0.5 & \lambda - 0.5 \end{vmatrix}$$

$$= \left[(\lambda - 0.2)^2 - 0.04 \right] \cdot \left[(\lambda - 0.5)^2 - 0.25 \right]$$

3. $\lambda_2 = 0$.

L matrix $\rightarrow L v = \lambda_2 v$

$L v = 0$

$\begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}$

$v_1 = v_2$

$v_3 = v_4$

$\rightarrow$ Node 1, 2

$\rightarrow$ Node 3, 4

用特征值百分

XP $\lambda_1 = \lambda_2 = 0$

$\lambda_1, \lambda_2, \lambda_3, \lambda_4$

$0, 0, 0.4, 1$

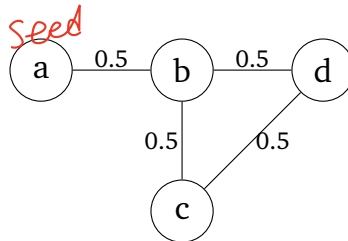
¹<https://en.wikipedia.org/wiki/Determinant>

4 Influence Spread

As we discussed the influence spread on directed graphs in the lecture. However, it is very straightforward to be applied to undirected graphs. Given the undirected graph below, the number on each edge denotes the probability the edge is live. Let the seed set $S = \{a\}$. We aim to compute the influence spread under the Independent Cascade (IC) model by taking the expectation over the deterministic graphs.

Let $X = \{X_{ab}, X_{bc}, X_{bd}, X_{cd}\}$ be the binary label for the edges where live=1 and blocked=0. Complete the following table and calculate the influence spread of this undirected graph:

(W3.2 - Slide 13 & 14)



$$\lambda - 0.2 = \pm 0.2$$

$$\lambda = 0.2 \pm 0.2$$

$$\lambda = 0, 0.4$$

X_{ab}	X_{bc}	X_{bd}	X_{cd}	$\Pr[X]$	# nodes reachable from S , $\sigma^X(S)$
0	0	0	0		1
0	0	0	1		1
0	0	1	0		1
0	0	1	1	$1 \sim 0.5$	1
0	1	0	0		1
0	1	0	1		1
0	1	1	0		1
0	1	1	1		1
1	0	0	0	$0.5 \cdot 0.5$	2
1	0	0	1	1	2
1	0	1	0		3
1	0	1	1	1	4
1	1	0	0		3
1	1	0	1		4
1	1	1	0		4
1	1	1	1		4